

Salesforce Project Documentation

WhatNext Vision Motors CRM

Prepared for SmartBridge Project Review
Salesforce Platform Developer Project

Contents

Section	Description
1	Project overview and purpose
2	Objectives and current implementation scope
3	Requirement analysis and Salesforce data model
4	Salesforce configuration and Lightning app setup
5	Process automation using record triggered flows
6	Apex trigger and trigger handler implementation
7	Batch Apex and Scheduled Apex
8	User interface and sample records
9	Functional testing evidence
10	Conclusion

This document describes only the current Salesforce implementation created for the WhatNext Vision Motors project. It excludes demo-video notes, proposed future improvements, and unrelated project material.

1. Project Overview

WhatNext Vision Motors is a Salesforce CRM implementation for an automotive sales and service workflow. The build organizes vehicle inventory, dealer information, customer records, vehicle orders, test-drive bookings, and service requests inside Salesforce so that the core business process can be managed from one Lightning application.

The project focuses on three practical business controls. First, the ordering process assigns a suitable dealer to a vehicle order using customer and dealer location information. Second, the system prevents confirmed orders when the selected vehicle does not have available stock. Third, the test-drive process sends reminder communication so customers receive timely appointment information.

The implementation uses standard Salesforce configuration, custom objects, custom fields, lookup relationships, Lightning app setup, record-triggered flows, Apex trigger logic, a trigger handler class, and asynchronous Apex processing.

2. Objectives and Current Implementation Scope

2.1 Project Objectives

- Create a Salesforce CRM structure for vehicle, dealer, customer, order, test-drive, and service-request management.
- Maintain vehicle stock information and use it while processing customer orders.
- Assign vehicle orders to a dealer based on the customer address and dealer location stored in Salesforce.
- Restrict order placement when the selected vehicle is out of stock.
- Send an automatic email reminder for scheduled test drives.
- Use Apex, triggers, trigger handlers, Batch Apex, and Scheduled Apex to demonstrate platform development concepts.

2.2 Current Scope

The current scope is the completed Salesforce build that can be reviewed in the developer org. It includes configured custom objects, fields, tabs, a Lightning app, record pages, sample data, flows, Apex trigger code, batch-processing logic, scheduled execution design, and functional testing evidence.

Dealer assignment is implemented with Salesforce record data. The flow compares stored customer address information with dealer location information and writes the selected dealer back to the order record. This represents the project requirement inside the current org without depending on an external maps or geocoding service.

Area	Included Implementation	Review Evidence
Data model	Custom objects, fields, picklists, auto number fields, and lookup fields	Object Manager field screens
Ordering	Vehicle Order linked with customer, vehicle, and assigned dealer	Order and validation screens
Inventory	Stock Quantity and Status used by automation and Apex	Vehicle and error screens
Dealer assignment	Flow retrieves customer and dealer records, then updates the order	Flow Builder screens
Test drive	Scheduled path sends reminder email before the appointment	Flow and Gmail evidence
Development	Trigger, handler, Batch Apex, and Scheduled Apex logic	Developer Console screens

3. Requirement Analysis and Salesforce Data Model

The project requires a data model that separates master records from transaction records. Vehicle, customer, and dealer records act as reusable master data. Vehicle orders, test drives, and service requests store operational activity and reference the master records through lookup fields.

Custom Object	Main Fields	Business Purpose
Vehicle	Name, Model, Price, Status, Stock Quantity, Dealer	Stores inventory and availability for each vehicle.
Vehicle Dealer	Name, Dealer Code, Location, Email, Phone	Stores dealer contact and location information.
Vehicle Customer	Name, Email, Phone, Address, Preferred Vehicle Type	Stores customer contact and preference data.
Vehicle Order	Order Number, Order Date, Status, Vehicle, Customer, Assigned Dealer	Captures order details and fulfillment status.
Vehicle Test Drive	Test Drive Name, Date, Status, Vehicle, Customer	Tracks appointments and reminder processing.
Vehicle Service Request	Request Name, Issue, Service Date, Status, Vehicle, Customer	Captures vehicle service and support requests.

The screenshot shows the 'Custom Object Definition Edit' interface in Salesforce. At the top, there are buttons for 'Save', 'Save & New', and 'Cancel'. Below this is the 'Custom Object Information' section, which includes a warning about changing labels and a 'Required Information' indicator. The form contains the following fields:

- Label:** 'Vehicle' (with an example of 'Account')
- Plural Label:** 'Vehicles' (with an example of 'Accounts')
- Starts with vowel sound:** An unchecked checkbox.
- Object Name:** 'Vehicle' (with an example of 'Account')
- Description:** A large text area.
- Context-Sensitive Help Setting:** Two radio buttons; the first is selected ('Open the standard Salesforce.com Help & Training window').
- Content Name:** A dropdown menu set to 'None'.

Below the 'Custom Object Information' section is the 'Enter Record Name Label and Format' section, which includes a note about the Record Name appearing in various Salesforce contexts.

Figure 1 Creating the Vehicle custom object in Salesforce

SETUP > OBJECT MANAGER Vehicle					
Details	Fields & Relationships 9 Items, Sorted by Field Label Quick Find New Deleted Fields Field Dependencies Set History Tracking				
Fields & Relationships	FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Page Layouts	Created By	CreatedById	Lookup(User)		
Lightning Record Pages	Last Modified By	LastModifiedById	Lookup(User)		
Buttons, Links, and Actions	Owner	OwnerId	Lookup(User,Group)		✓
Compact Layouts	Price	Price__c	Currency(18, 0)		▼
Field Sets	Status	Status__c	Picklist		▼
Object Limits	Stock Quantity	Stock_Quantity__c	Number(18, 0)		▼
Record Types	Vehicle Dealer	Vehicle_Dealer__c	Lookup(Vehicle Dealer)		✓ ▼
Related Lookup Filters	Vehicle Model	Vehicle_Model__c	Picklist		▼
Search Layouts	Vehicle Name	Name	Text(80)		✓ ▼
List View Button Layout					
Restriction Rules					
Scoping Rules					

Figure 2 Vehicle object fields and relationships including stock and dealer lookup

SETUP > OBJECT MANAGER Vehicle Order					
Details	Fields & Relationships 8 Items, Sorted by Field Label Quick Find New Deleted Fields Field Dependencies Set History Tracking				
Fields & Relationships	FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Page Layouts	Created By	CreatedById	Lookup(User)		
Lightning Record Pages	Last Modified By	LastModifiedById	Lookup(User)		
Buttons, Links, and Actions	Order Date	Order_Date__c	Date		▼
Compact Layouts	Owner	OwnerId	Lookup(User,Group)		✓
Field Sets	Status	Status__c	Picklist		▼
Object Limits	Vehicle	Vehicle__c	Lookup(Vehicle)		✓ ▼
Record Types	Vehicle Customer	Vehicle_Customer__c	Lookup(Vehicle Customer)		✓ ▼
Related Lookup Filters	Vehicle Order Number	Name	Auto Number		✓ ▼
Search Layouts					
List View Button Layout					
Restriction Rules					
Scoping Rules					

Figure 3 Vehicle Order fields including order date status customer vehicle and order number

SETUP > OBJECT MANAGER

Vehicle Customer

Details	Fields & Relationships 8 Items, Sorted by Field Label	Q Quick Find	New	Deleted Fields	Field Dependencies	Set History Tracking
Fields & Relationships						
Page Layouts						
Lightning Record Pages						
Buttons, Links, and Actions						
Compact Layouts						
Field Sets						
Object Limits						
Record Types						
Related Lookup Filters						
Search Layouts						
List View Button Layout						
Restriction Rules						

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Address	Address__c	Text(50)		
Created By	CreatedById	Lookup(User)		
Email	Email__c	Email		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Phone	Phone__c	Phone		
Preferred Vehicle Type	Preferred_Vehicle_Type__c	Picklist		
Vehicle Customer Name	Name	Text(80)		✓

Figure 4 Vehicle Customer fields used for customer contact and address details

SETUP > OBJECT MANAGER

Vehicle Test Drive

Details	Fields & Relationships 8 Items, Sorted by Field Label	Q Quick Find	New	Deleted Fields	Field Dependencies	Set History Tracking
Fields & Relationships						
Page Layouts						
Lightning Record Pages						
Buttons, Links, and Actions						
Compact Layouts						
Field Sets						
Object Limits						
Record Types						
Related Lookup Filters						
Search Layouts						
List View Button Layout						
Restriction Rules						
Scoping Rules						

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Status	Status__c	Picklist		
Test Drive Date	Test_Drive_Date__c	Date		
Vehicle	Vehicle__c	Lookup(Vehicle)		✓
Vehicle Customer	Vehicle_Customer__c	Lookup(Vehicle Customer)		✓
Vehicle Test Drive Name	Name	Text(80)		✓

Figure 5 Vehicle Test Drive fields used for appointment and reminder processing

SETUP > OBJECT MANAGER

Vehicle Service Request

Details	Fields & Relationships 9 Items, Sorted by Field Label	Q Quick Find	New	Deleted Fields	Field Dependencies	Set History Tracking
Fields & Relationships						
Page Layouts						
Lightning Record Pages						
Buttons, Links, and Actions						
Compact Layouts						
Field Sets						
Object Limits						
Record Types						
Related Lookup Filters						
Search Layouts						
List View Button Layout						
Restriction Rules						
Scoping Rules						

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Issue Description	Issue_Description__c	Text(60)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Service Date	Service_Date__c	Date		
Status	Status__c	Picklist		
Vehicle	Vehicle__c	Lookup(Vehicle)		✓
Vehicle Customer	Vehicle_Customer__c	Lookup(Vehicle Customer)		✓
Vehicle Service Request Name	Name	Text(80)		✓

Figure 6 Vehicle Service Request fields used for support and service tracking

4. Salesforce Configuration and Lightning App Setup

The project uses a dedicated Lightning app so that all project objects are available from one navigation workspace. Custom tabs were created for the core objects and then added to the application navigation during setup.

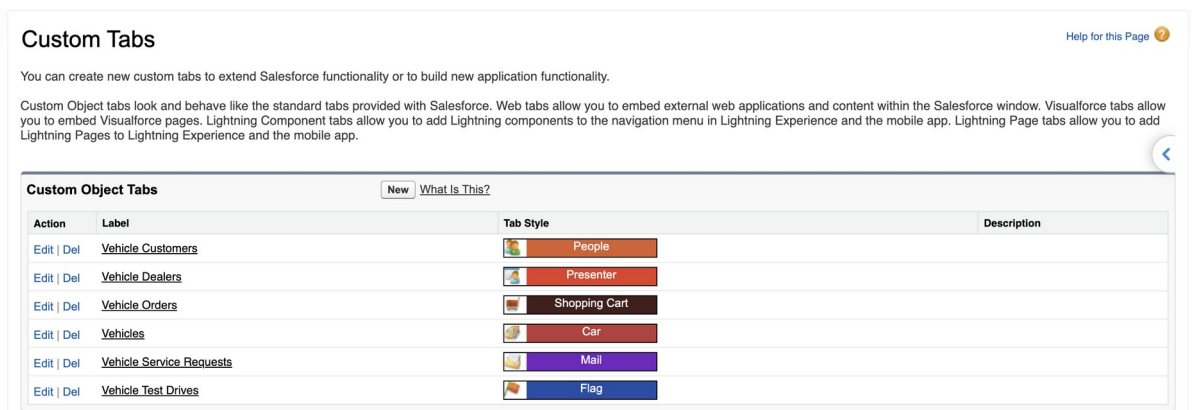


Figure 7 Custom tabs created for the project objects

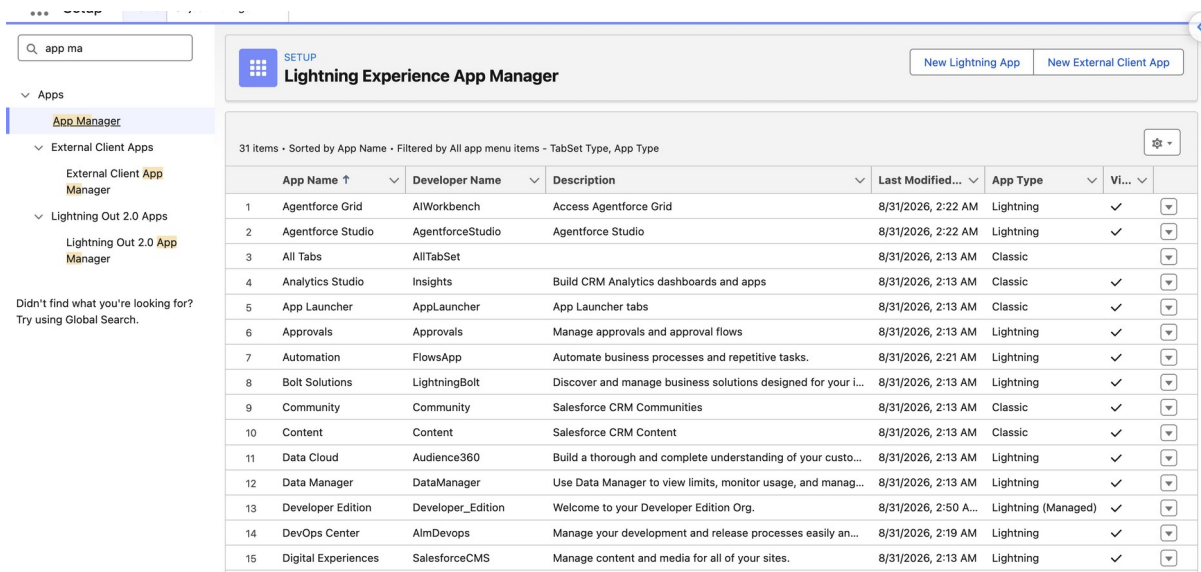


Figure 8 Lightning Experience App Manager showing the custom application

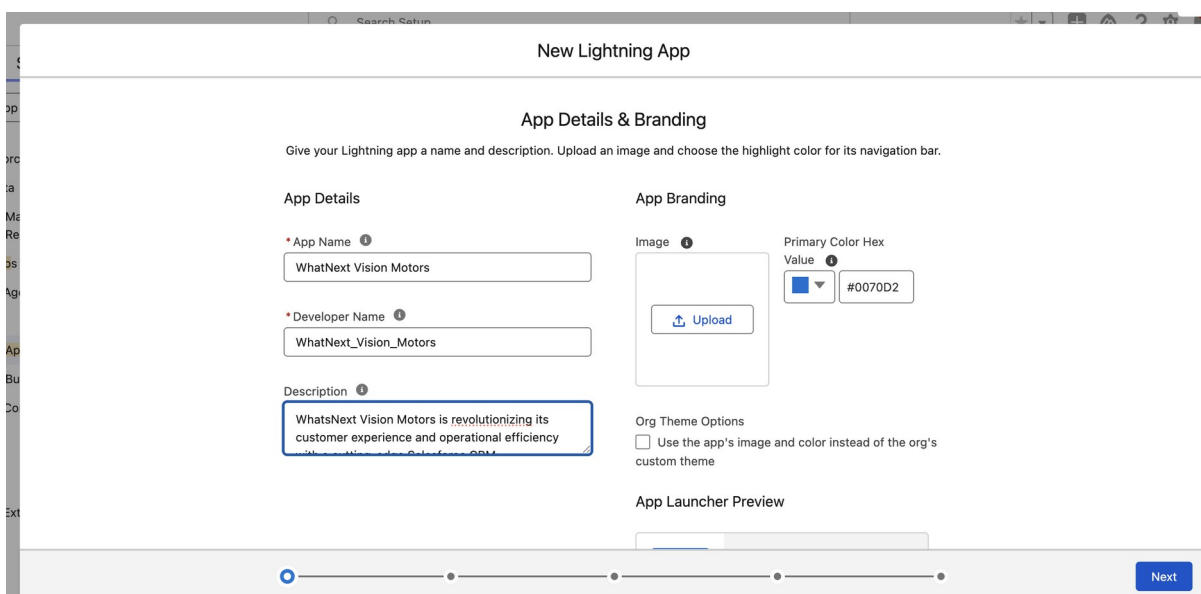


Figure 9 WhatNext Vision Motors Lightning app details and branding

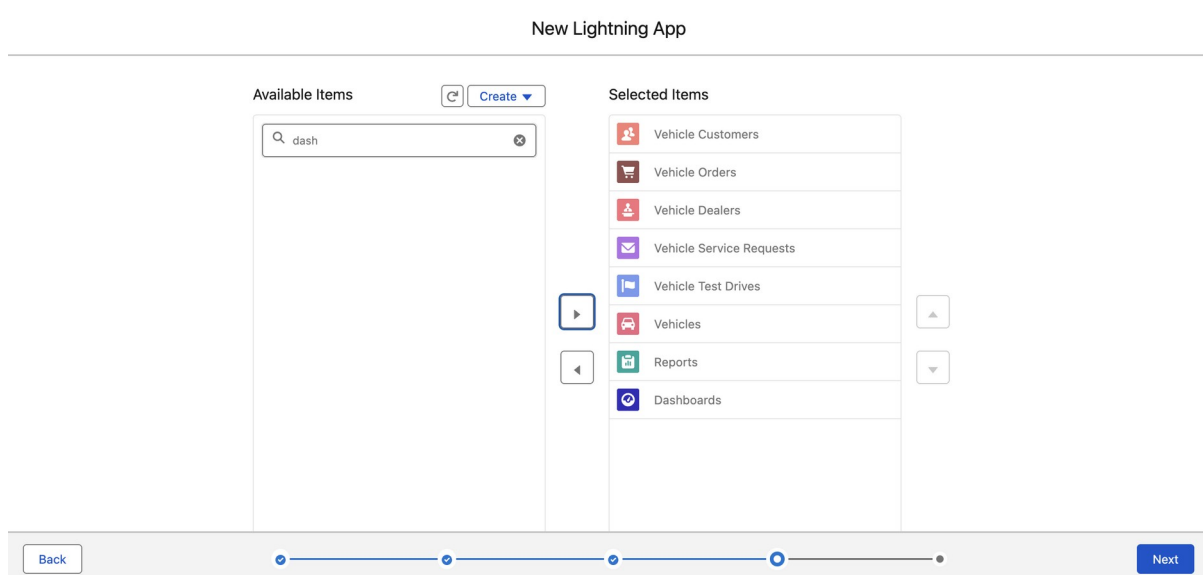


Figure 10 Navigation items selected for the WhatNext Vision Motors app

5. Process Automation Using Record Triggered Flows

5.1 Auto Assign Dealer Flow

The Auto Assign Dealer flow starts when a Vehicle Order record is created or updated. The entry condition focuses on pending orders, then the flow retrieves the related customer record, finds a matching dealer record using location information, and updates the order with the assigned dealer.

Flow Step	Configuration	Purpose
Start	Runs on Vehicle Order when the status meets the entry condition	Starts assignment at the correct order stage.
Get Customer Information	Retrieves the customer linked to the order	Reads the address used for matching.
Get Nearest Dealer	Retrieves a dealer using the location criteria	Finds the matching dealer record.
Assign Dealer to Order	Updates the order with the selected dealer	Prepares the order for fulfillment.

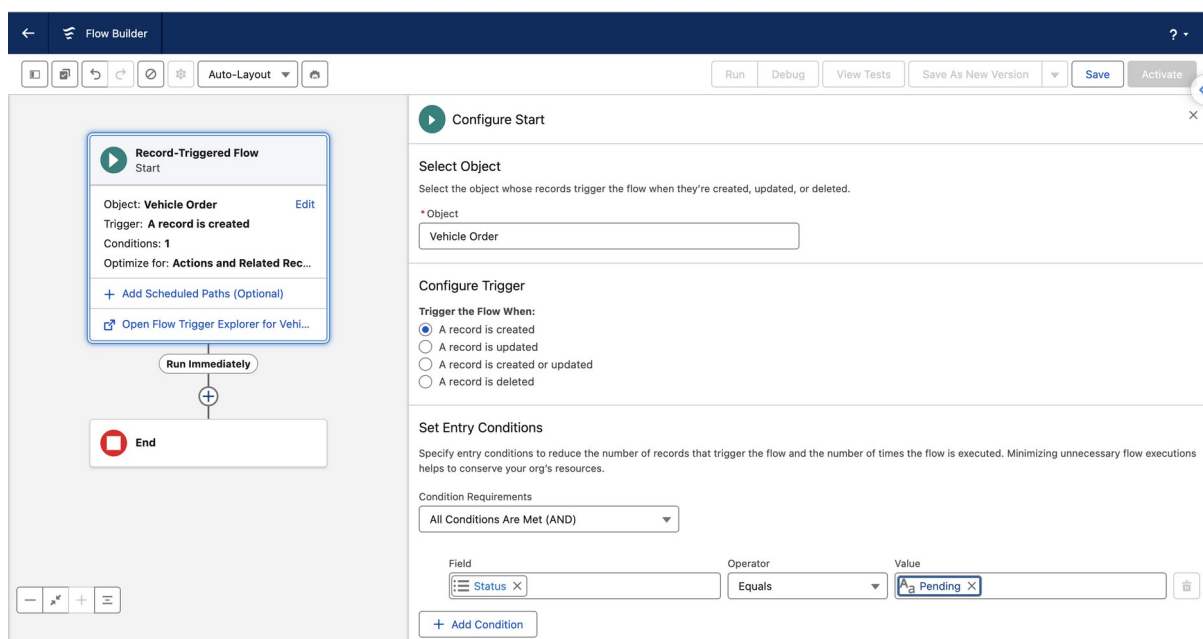


Figure 11 Auto Assign Dealer flow start condition on Vehicle Order

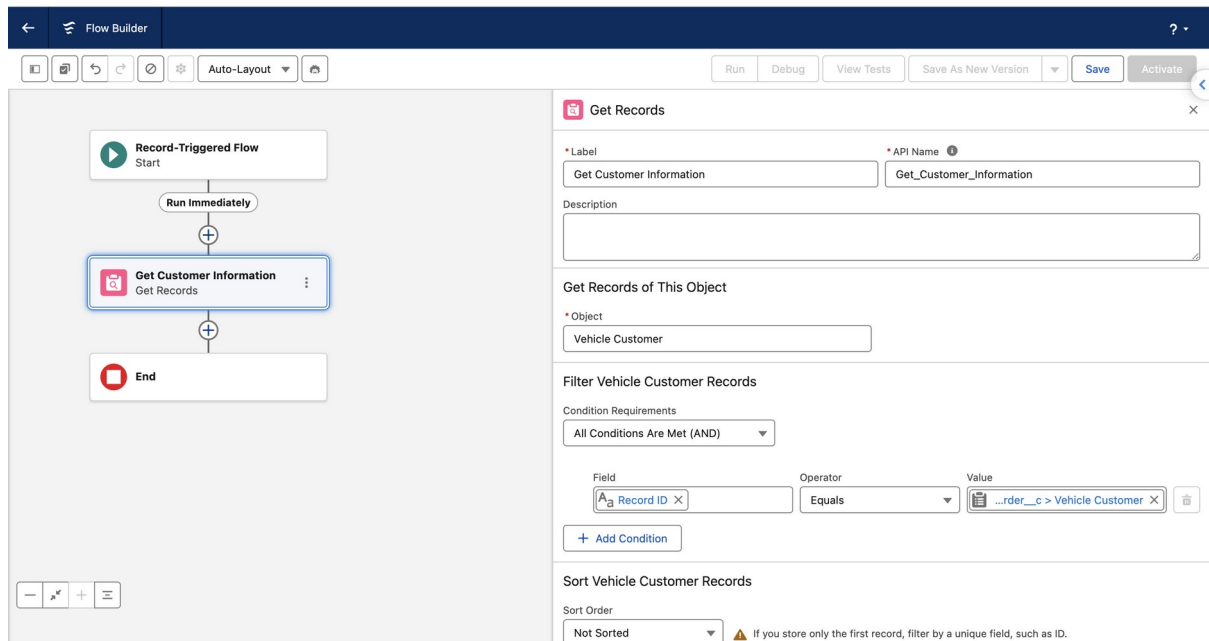


Figure 12 Flow step that retrieves the related customer information

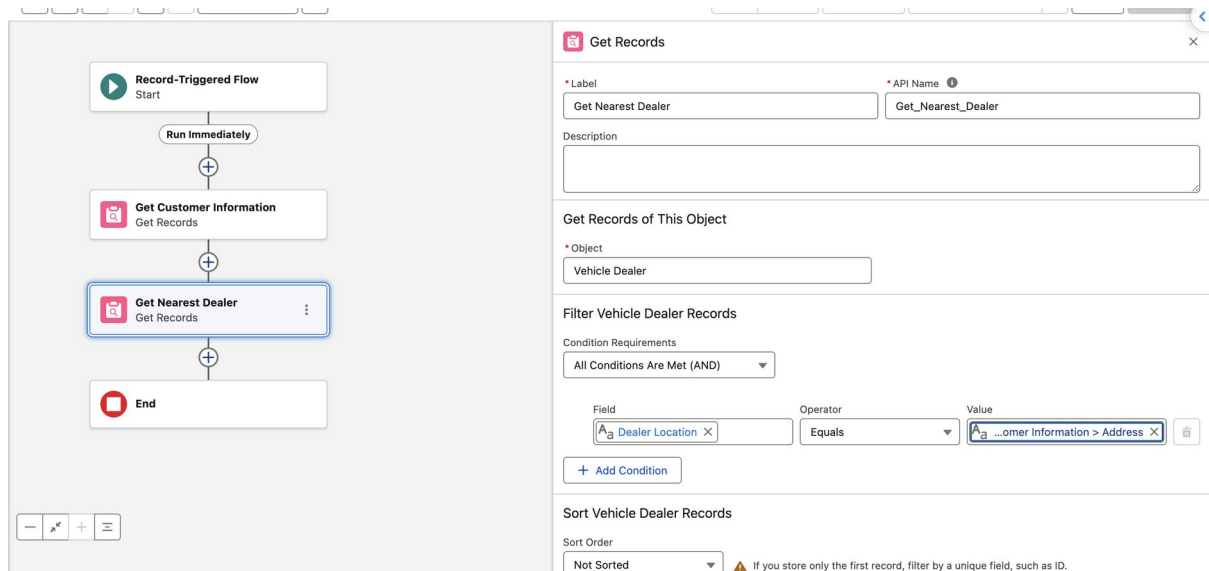


Figure 13 Flow step that retrieves the matching dealer record

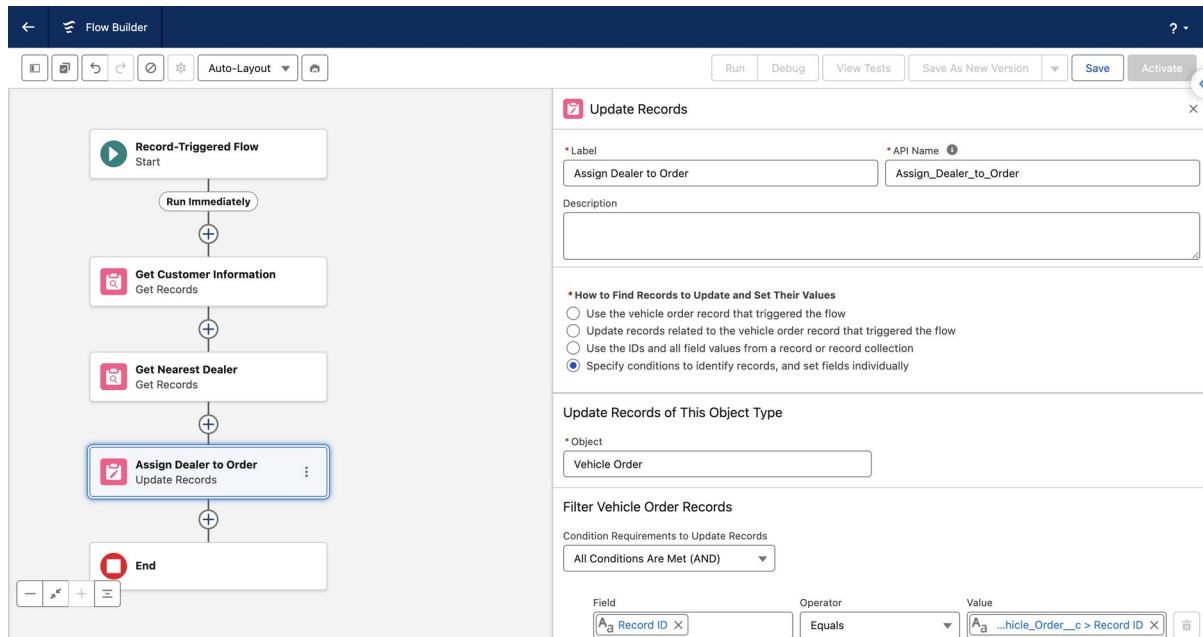


Figure 14 Flow update step that writes the assigned dealer to the order

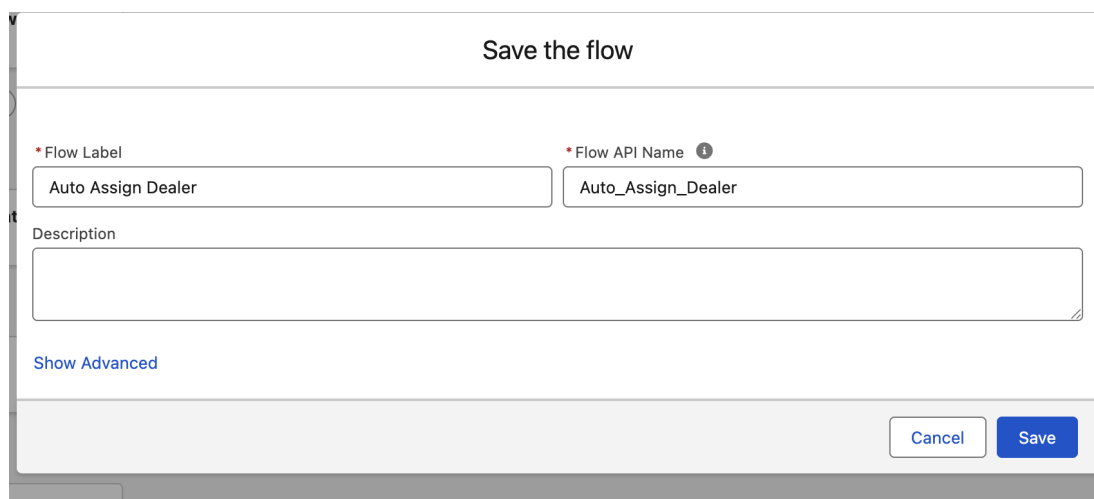


Figure 15 Saved Auto Assign Dealer flow

5.2 Test Drive Reminder Flow

The test-drive reminder automation uses a scheduled path on Vehicle Test Drive records. Before the appointment date, the flow retrieves the customer details and sends an email reminder. This supports the customer communication requirement without manual follow-up.

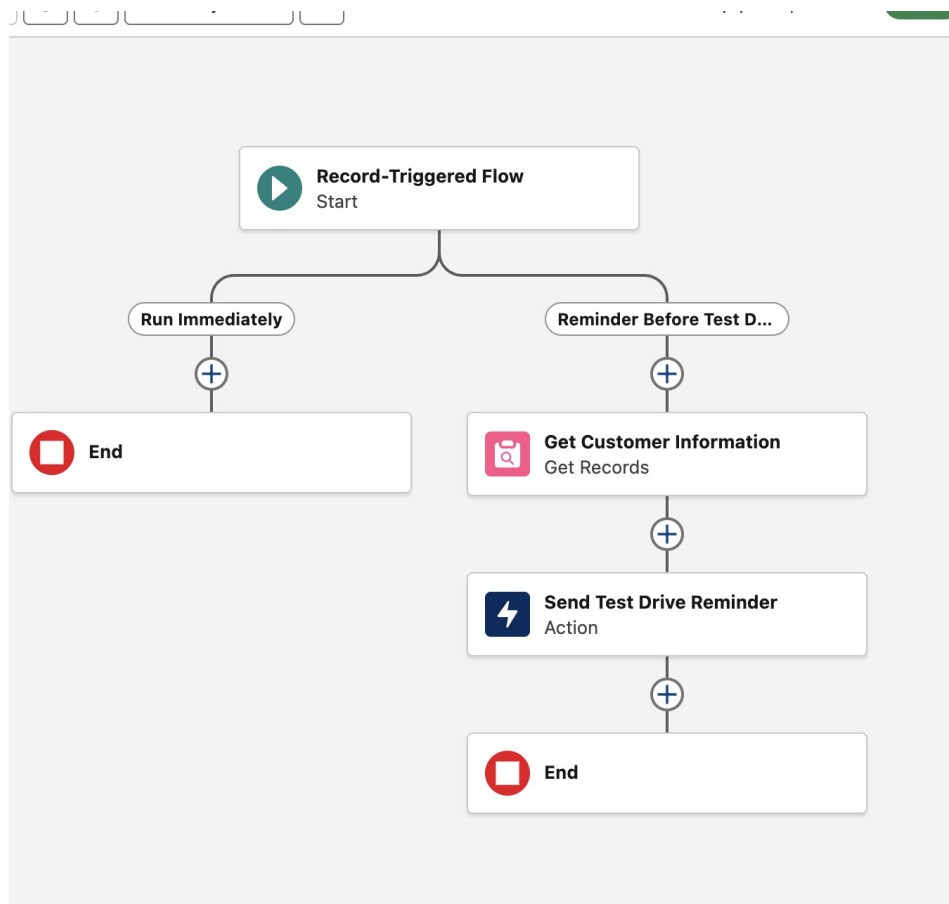


Figure 16 Test-drive reminder flow with scheduled reminder path

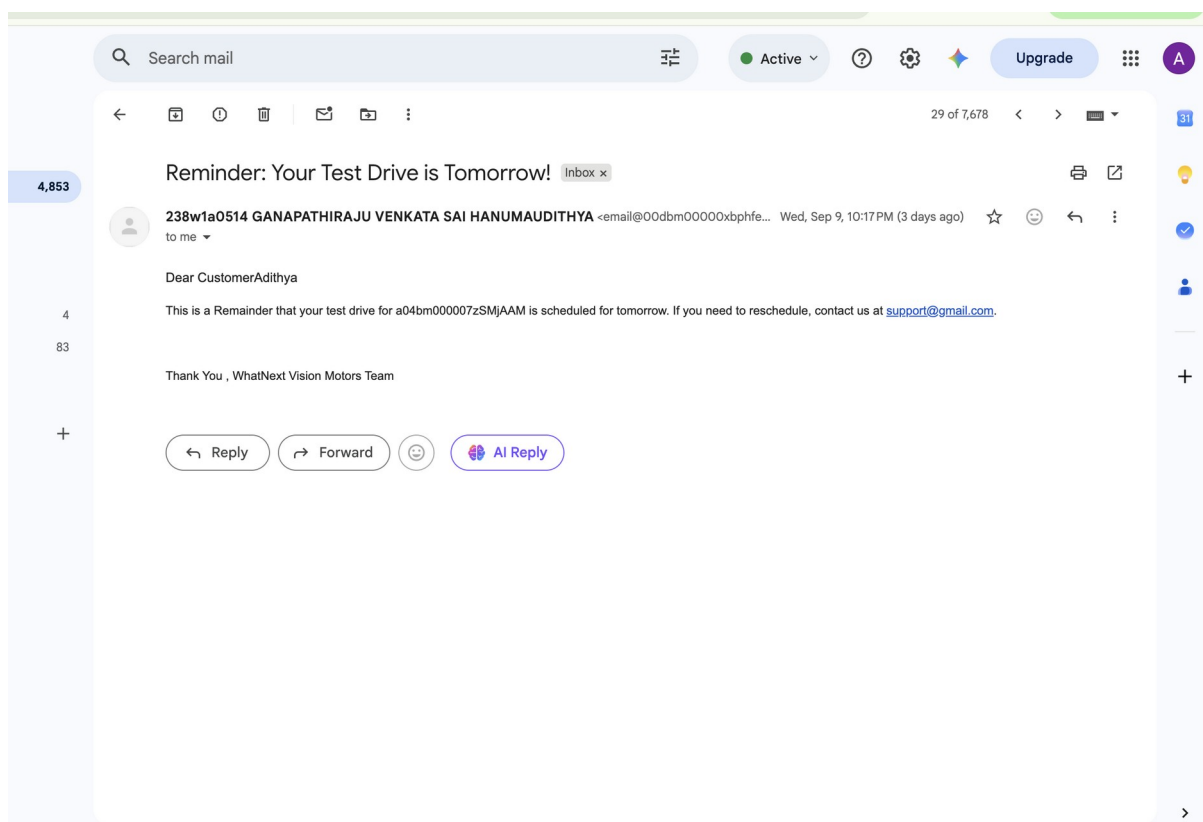


Figure 17 Reminder email received for a scheduled test drive

6. Apex Trigger and Trigger Handler Implementation

Apex is used for stock validation because order confirmation is a business rule that should be enforced consistently at the platform level. The trigger is kept small and delegates logic to a handler class. This follows Salesforce best practice by keeping trigger code readable and moving business logic into a reusable class.

Apex Artifact	Responsibility	Result
VehicleOrderTrigger	Runs on insert and update for Vehicle Order	Passes context to the handler.
VehicleOrderTriggerHandler	Checks related vehicle stock and order status	Blocks unavailable vehicle orders.
Stock validation	Uses Vehicle and Stock Quantity values	Shows an error for out-of-stock orders.

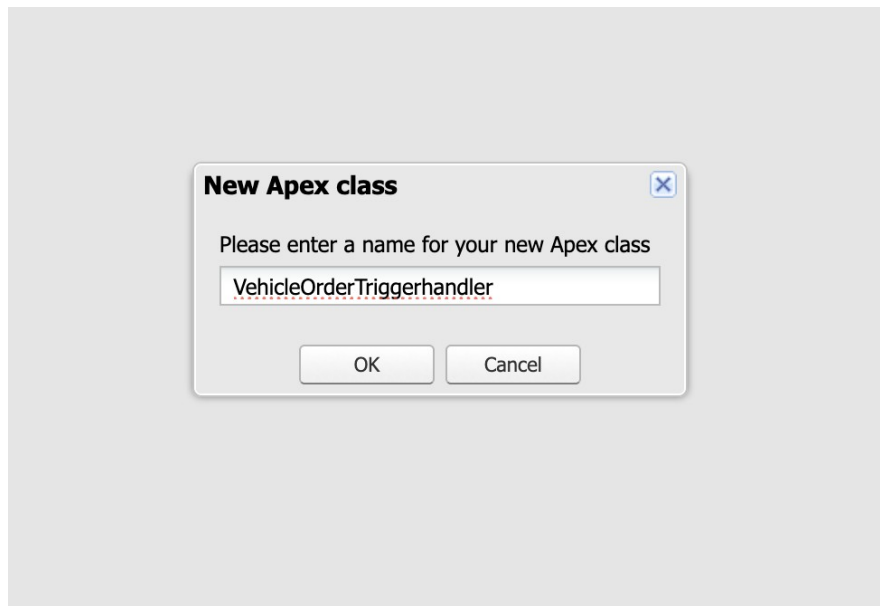


Figure 18 Creating the VehicleOrderTriggerHandler Apex class

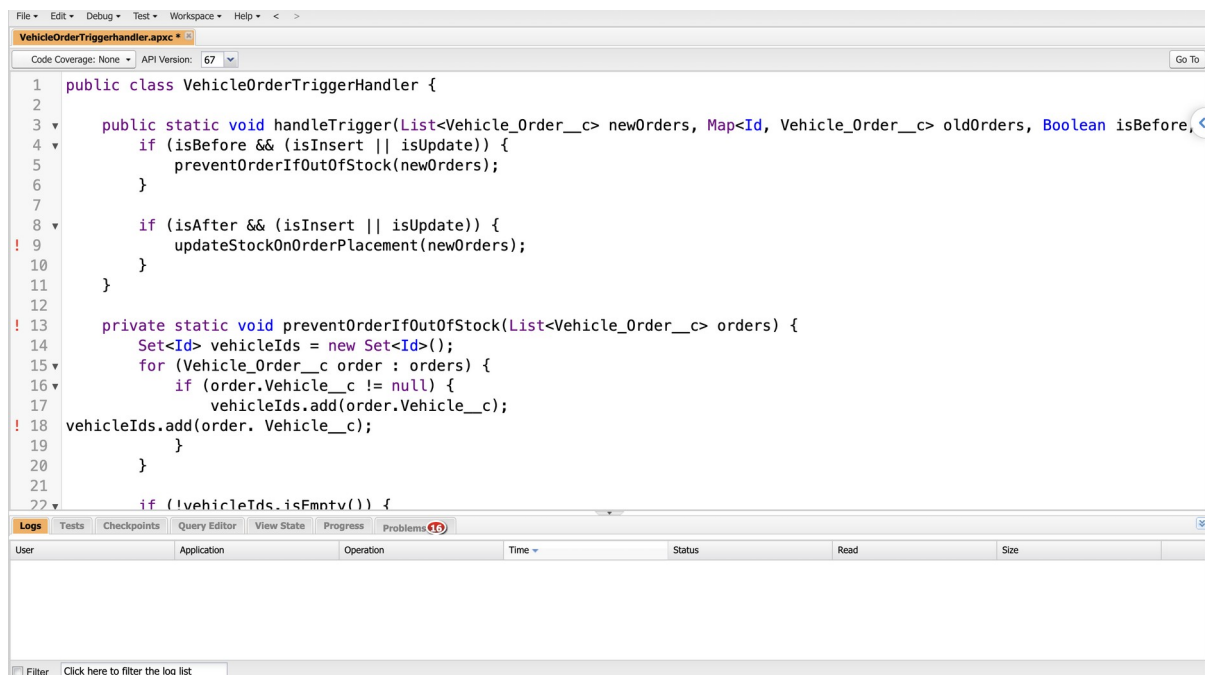


Figure 19 VehicleOrderTriggerHandler class in Developer Console

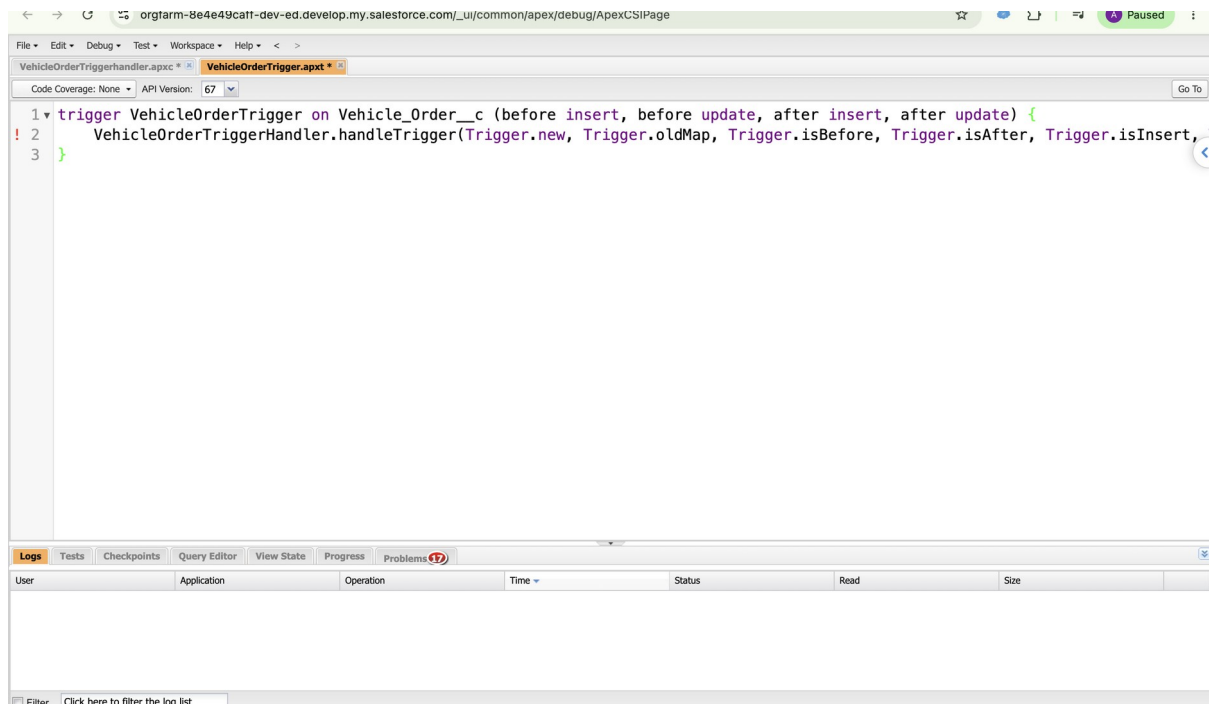


Figure 20 VehicleOrderTrigger delegating logic to the handler

Trigger pattern used in the project:

```

trigger VehicleOrderTrigger on Vehicle_Order__c (before insert, before update, after insert,
after update) {
    VehicleOrderTriggerHandler.handleTrigger(trigger.new, trigger.oldMap, trigger.isBefore,
trigger.isAfter, trigger.isInsert, trigger.isUpdate);
}

```

7. Batch Apex and Scheduled Apex

Batch Apex is used to process multiple pending orders in manageable groups. Scheduled Apex is used to run the batch automatically. In this project, the batch checks pending orders against related vehicle stock and updates order status based on availability.

Class or Process	Role in the Project	Expected Status Update
VehicleOrderBatch	Queries pending orders and checks vehicle stock	Confirmed if stock is available; Pending if not.
VehicleOrderBatchScheduler	Runs the batch class on schedule	Keeps bulk order statuses updated.
Email notification logic	Supports automated processing communication	Improves order-handling visibility.

This asynchronous design is suitable for bulk records because large order updates can be processed outside a single user transaction and can be scheduled according to business needs.

8. User Interface and Sample Records

Sample records were created to verify that users can work with the configured objects from the Lightning application. The screenshots below show customer, dealer, vehicle, and order records used during testing.



Vehicle Customer
Adithya

Related

Details

Vehicle Customer Name

Adithya

Email

adithya.gvsh@gmail.com

Phone

(903) 070-0101

Address

Vijayawada

Preferred Vehicle Type

SUV

Created By


238w1a0514 GANAPATHIRAJU VENKATA SAI HANUMAUDITHYA,
9/6/2026, 3:04 AM

Owner


238w1a0514 GANAPATHIRAJU VENKATA SAI HANUMAUDITHYA

Last Modified By


238w1a0514 GANAPATHIRAJU VENKATA SAI HANUMAUDITHYA,
9/6/2026, 3:04 AM

Figure 21 Vehicle Customer sample record with email phone address and preferred vehicle type



Vehicle Dealer
Akhil

Related

Details

Vehicle Dealer Name

Akhil

Dealer Location

Vijayawada

Dealer Code

D-0001

Phone

(987) 868-6612

Email

akhilbabu@gmail.com

Created By


238w1a0514 GANAPATHIRAJU VENKATA SAI HANUMAUDITHYA,
9/6/2026, 3:07 AM

Owner


238w1a0514 GANAPATHIRAJU VENKATA SAI HANUMAUDITHYA

Last Modified By


238w1a0514 GANAPATHIRAJU VENKATA SAI HANUMAUDITHYA,
9/6/2026, 3:07 AM

New Contact

Edit

New Opportunity

Figure 22 Vehicle Dealer sample record with dealer location and contact details

Related

Details

Vehicle Name

Ferrari

Vehicle Model

SUV

Stock Quantity

100

Price

\$90,000

Vehicle Dealer

Akhil

Status

Available

Created By


238w1a0514 GANAPATHIRAJU VENKATA SAI HANUMAUDITHYA,
9/6/2026, 3:10 AM

Owner


238w1a0514 GANAPATHIRAJU VENKATA SAI HANUMAUDITHYA

Last Modified By


238w1a0514 GANAPATHIRAJU VENKATA SAI HANUMAUDITHYA,
9/6/2026, 3:10 AM

Figure 23 Vehicle sample record with model price stock quantity status and dealer

9. Functional Testing Evidence

The testing approach was based on direct Salesforce UI validation. Records were created and updated to confirm that object configuration, automation, Apex validation, and email reminders behaved as expected.

Test Case	Action Performed	Expected Result	Actual Result
Custom app access	Opened the custom Lightning app	Project tabs should appear	All project tabs displayed.
Data model setup	Checked Object Manager fields	Objects should have required fields	Fields were configured.
Dealer assignment	Created or updated a pending order	Flow should assign dealer	Order was updated by flow.
Stock validation	Saved a confirmed order with no stock	Order should be blocked	Out-of-stock error appeared.
Test-drive reminder	Scheduled a test drive	Customer should get email	Reminder email was received.
Apex structure	Reviewed trigger and handler	Trigger should call handler	Handler delegation was present.
Bulk processing	Reviewed batch and scheduler	Pending orders should update	Batch design supports updates.

New Vehicle Order

* = Required Information

Information

Vehicle Order Number

Owner
238w1a0514 GANAPATHIRAJU VENKATA SAI HANUMAUDITHYA

Vehicle Customer
Adithya

Vehicle
Ferrari

Order Date
9/10/2026

Status
Confirmed

Assigned Dealer
Search Vehicle Dealers...

We hit a snag.

Review the errors on this page.

- This vehicle is out of stock. Order cannot be placed.

Cancel Save & New Save

Figure 26 Stock validation error when an out-of-stock vehicle is selected for a confirmed order

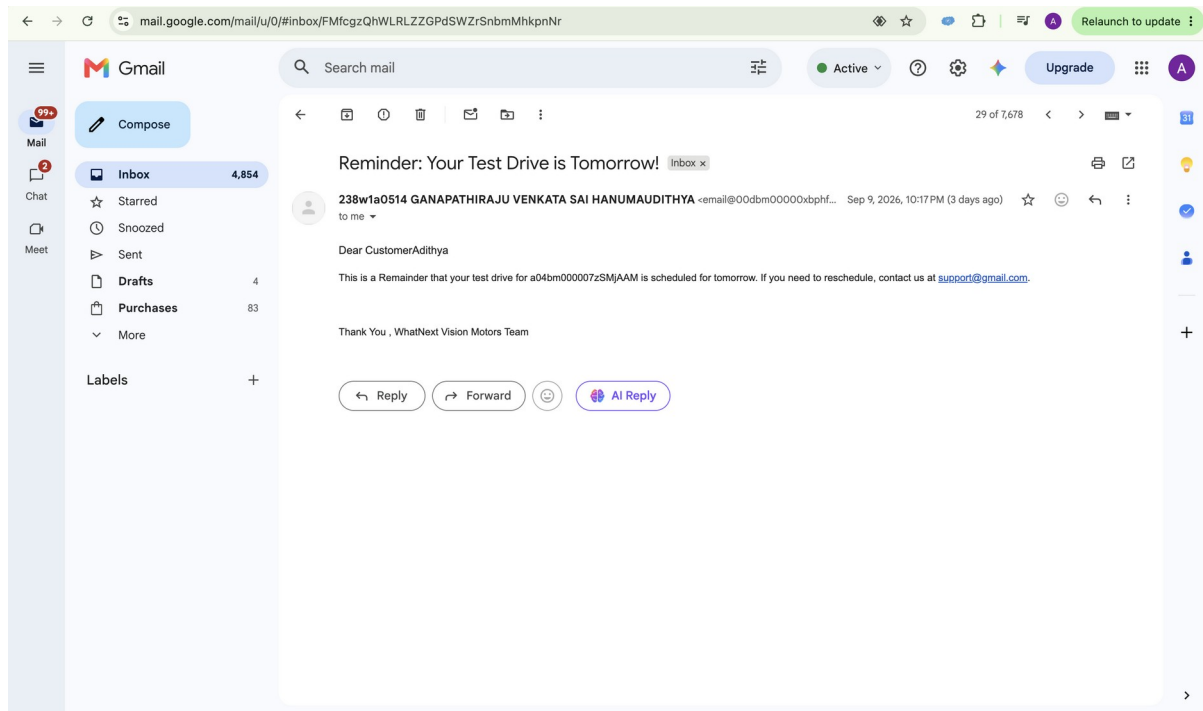


Figure 27 Gmail evidence for the automated test-drive reminder

10. Conclusion

The WhatNext Vision Motors Salesforce project is a complete CRM implementation for the current review scope. It stores automotive master data and transaction records, provides a dedicated Lightning app, automates dealer assignment, prevents invalid out-of-stock orders, sends test-drive reminders, and demonstrates Apex trigger, handler, batch, and scheduled Apex concepts.

The screenshots included in this document show the implementation from setup through testing, giving the SmartBridge reviewer clear evidence of the completed Salesforce configuration and development work.